batmonitor 0.74-py: decision tables (c/bm_tables.h, 15 tables, 67 rows) - first matching row wins, - = any; green 1, red 0; share = input combinations the row decides (DEAD = never reached)

D7 house on AC-out1

D7								
mode_off	mode_on	over_on	under_off	under_long	on	**on**	**under**	share
1	-	-	-	-	-	**0**	**0**	32/64
-	1	-	-	-	-	**1**	**0**	16/64
-	-	1	-	-	-	**1**	**0**	8/64
-	-	-	1	1	-	**0**	**1**	2/64
-	-	-	1	-	1	**1**	**1**	1/64
-	-	-	1	-	-	**0**	**1**	1/64
-	-	-	-	-	-	**KEEP**	**0**	4/64

D7: house on AC-out1. HOUSE_ACOUT 0 = off, 1 = on, auto: on above 99 W, off after 60 s below 30 W.

D5 DO4 pulse

D5						
lockout_over	acout_over	pcc_over	grace_over	pcc_hard	**do4**	share
0	-	-	-	-	**0**	16/32
-	1	-	-	-	**1**	8/32
-	-	1	1	-	**1**	2/32
-	-	1	-	1	**1**	1/32
-	-	-	-	-	**0**	5/32

D5: DO4 pulse (FoxESS shedding): PCC > 20 kW for the grace time or > DO4_HARD_W, or grid offline and PV overfeeds one AC-out; never twice within the lockout.

D3 discharge PROP / IDLE

D3						
any_dis	house_import	prop_prev	prop_hold	deficit_gt_hold	**prop**	share
0	-	-	-	-	**0**	16/32
-	1	-	-	-	**1**	8/32
-	-	1	-	1	**1**	2/32
-	-	-	1	1	**1**	1/32
-	-	-	-	-	**0**	5/32

D3: PROPORTIONAL (1) while importing, or (already proportional / night floor) while the deficit > 20 W.

SoC balancing

DLEAD					
both_ok	has_lead	diff_gt_on	diff_lt_off	**lead**	share
0	-	-	-	**NONE**	8/16
-	0	1	-	**NEW**	2/16
-	1	-	1	**END**	2/16
-	-	-	-	**KEEP**	4/16

SoC balancing: the stack more than SOC_BALANCE_ON ahead leads, cleared below SOC_BALANCE_OFF.

D1 Sofar first, source, winter heat pump rule

D1_LATCH				
b1_above_min	bat1_first	b1_ge_release	**bat1_first**	share
0	-	-	**0**	4/8
1	0	1	**1**	1/8
-	-	-	**KEEP**	3/8

D1: Sofar first latch. Off at or below BAT1_SOC_MIN, on again from BAT1_SOC_MIN + 2 %.

bat1_first

D1_B1_FIRST			
bat1_first	wp_on	**b1_first**	share
1	1	**1**	1/4
-	-	**0**	3/4

D1: Sofar first acts only while the heat pump runs.

b1_first

D1_SRC					
night	b1_first	winter	wp_on	**src**	share
0	-	-	-	**DAY**	8/16
1	1	-	-	**B1**	4/16
1	-	1	1	**WPG**	1/16
-	-	-	-	**STK**	3/16

D1: S3 source of the discharge. DAY by day, B1 = Sofar first at night, WPG = winter night with the heat pump running (no night floor), STK = night floor.

D1_WP_CAP				
winter	wp_on	b1_dis	**wp_cap**	share
1	1	1	**ZERO**	1/8
1	1	0	**MAX**	1/8
-	-	-	**NONE**	6/8

D1: winter heat pump rule. Stacks give 0 W while the Sofar Bat1 discharges, else at most WP_WINTER_W.

D2 per stack: latches and mode

D2_CHARGE_MODE				
surplus_gt_on	chg_prev	surplus_gt_hold	**charge_mode**	share
1	-	-	**1**	4/8
-	1	1	**1**	1/8
-	-	-	**0**	3/8

D2: charge mode on the surplus (> 200 W), held down to 20 W once charging.

charge_mode

D2_MODE													
bms_ok	force	low	stale	charge_mode	block	soc_lt_100	ff	top	prot	charging	**mode**	share
0	-	-	-	-	-	-	-	-	-	-	**BMS_MISSING**	1024/2048
-	1	-	-	-	-	-	-	-	-	-	**FORCE**	512/2048
-	-	1	-	-	-	-	-	-	-	-	**FORCE**	256/2048
-	-	-	1	-	-	-	-	-	-	-	**STALE**	128/2048
-	-	-	-	1	1	-	-	-	-	-	**LADESPERRE**	32/2048
-	-	-	-	1	-	1	-	-	-	-	**CHARGE**	16/2048
-	-	-	-	1	-	-	1	-	-	-	**CHARGE**	8/2048
-	-	-	-	1	-	-	-	1	-	-	**CHARGE**	4/2048
-	-	-	-	1	-	-	-	-	-	-	**FULL**	4/2048
-	-	-	-	-	-	-	1	-	-	-	**HOLD**	32/2048
-	-	-	-	-	-	-	-	-	1	-	**PROT**	16/2048
-	-	-	-	-	-	-	-	-	-	1	**CHARGING**	8/2048
-	-	-	-	-	-	-	-	-	-	-	**DIS**	8/2048

D2: mode per stack, priority from top to bottom (force / low / prot = the latches above, ff = force_full_from_pv names this stack, top = top hold window of this stack).

mode

D2_PROT				
bms_ok	soc_lt_min	soc_ge_release	**prot**	share
1	1	-	**1**	2/8
1	-	1	**0**	1/8
-	-	-	**KEEP**	5/8

D2: SoC protection per stack, SOC_MIN .. SOC_MIN_RELEASE (only while the BMS reports).

prot

D2_FORCE				
bms_ok	soc_lt_force	soc_ge_force_release	**force**	share
1	1	-	**1**	2/8
1	-	1	**0**	1/8
-	-	-	**KEEP**	5/8

D2: grid force charge per stack, SOC_FORCE .. SOC_FORCE_RELEASE.

force

D2_LOW					
bms_ok	low_on	low_off	soc_ge_release	**low**	share
1	1	-	-	**1**	4/16
1	-	1	1	**0**	1/16
-	-	-	-	**KEEP**	11/16

D2: LOW force charge per stack: DCL 0 or lowest cell < CELL_FORCE_V; ends once DCL is back, the cell >= CELL_FORCE_RELEASE_V and the SoC >= SOC_MIN_RELEASE.

low

D4 summer peak window

D4_PEAK			
pcc_peak	new_day	**peak_today**	share
1	-	**1**	2/4
-	1	**0**	1/4
-	-	**KEEP**	1/4

D4: PCC above 20 kW seen today (display).

D4								
enabled	season	fc_cap_ahead	surplus_cap_on	chg_prev	surplus_cap_hold	**block**	**cap**	share
0	-	-	-	-	-	**0**	**0**	32/64
-	0	-	-	-	-	**0**	**0**	16/64
-	-	0	-	-	-	**0**	**0**	8/64
-	-	-	1	-	-	**0**	**1**	4/64
-	-	-	-	1	1	**0**	**1**	1/64
-	-	-	-	-	-	**1**	**1**	3/64

D4: summer peak window, outputs block (wait below EXPORT_CAP) and cap (charge only above it).

block

effect tables (bm_logic.py / bm_logic.c)

src

src_rule BM_SRC_RULE: the discharge per source				
src	b1_signed	trickle	idle_w	prop_hold
DAY	0	0	20	0
B1	0	0	20	0
STK	1	1	0	1
WPG	0	0	20	0

mode_rule BM_MODE_RULE: what a mode does			
mode	**action**	**capped**	**rule text**
BMS_MISSING	0 W	0	NAME
FORCE	FORCE_CHARGE_W	1	NAME(soc %)
STALE	0 W	0	NAME
LADESPERRE	0 W	0	NAME(peak h)
CHARGE	charge amount	1	NAME SRC:..
FULL	0 W	0	NAME
PROT	0 W	0	NAME(soc %)
CHARGING	0 W	0	NAME(BMS W)
DIS	discharge amount	1	(discharge path)
HOLD	0 W	0	NAME(soc %)

batmonitor: observed states, last 4 days (wagodb.pv_victron, one row per minute = the state of the last 5 s cycle; generated 2026-11-04 18:50)

coverage 01.11. 00:00 .. 04.11. 18:50: 5451 of 5451 minutes with a state (100.00 %)			
day	**minutes**	**D1 cells**	**rule states**
Sun 01.11.	1440	8	24
Mon 02.11.	1440	12	30
Tue 03.11.	1440	6	26
Wed 04.11.	1131	8	26
gaps (minutes without a row)			
none			

D1 cells (pv_victron.bm_d1_id) through the D1 tables of bm_tables.h					
cell	**bits**	**minutes**	**set bits**	**src**	**wp_cap**
95	1+2+4+8+16+64	1522	night, wp_on, b1_above_min, b1_ge_release, winter, bat1_first	B1	WP_WINTER_W
92	4+8+16+64	1150	b1_above_min, b1_ge_release, winter, bat1_first	DAY	none
93	1+4+8+16+64	974	night, b1_above_min, b1_ge_release, winter, bat1_first	STK	none
94	2+4+8+16+64	855	wp_on, b1_above_min, b1_ge_release, winter, bat1_first	DAY	WP_WINTER_W
87	1+2+4+16+64	362	night, wp_on, b1_above_min, winter, bat1_first	B1	WP_WINTER_W
85	1+4+16+64	232	night, b1_above_min, winter, bat1_first	STK	none
127	1+2+4+8+16+32+64	137	night, wp_on, b1_above_min, b1_ge_release, winter, b1_dis, bat1_first	B1	0 W
126	2+4+8+16+32+64	124	wp_on, b1_above_min, b1_ge_release, winter, b1_dis, bat1_first	DAY	0 W
124	4+8+16+32+64	45	b1_above_min, b1_ge_release, winter, b1_dis, bat1_first	DAY	none
84	4+16+64	44	b1_above_min, winter, bat1_first	DAY	none
86	2+4+16+64	4	wp_on, b1_above_min, winter, bat1_first	DAY	WP_WINTER_W
125	1+4+8+16+32+64	2	night, b1_above_min, b1_ge_release, winter, b1_dis, bat1_first	STK	none

rule states L1 | L2 | L3 (measured values in brackets removed), 53 different		
minutes	**%**	**L1 | L2 | L3**
1351	24.8	PROP SRC:B1 EQ WPMAX(100) | PROP SRC:B1 EQ WPMAX(100) | PROP SRC:B1 EQ WPMAX(100)
1113	20.4	PROP SRC:STK EQ | PROP SRC:STK EQ | PROP SRC:STK EQ
428	7.9	CHARGE SRC:DAY EQ | CHARGE SRC:DAY EQ | CHARGE SRC:DAY EQ
357	6.5	CHARGE SRC:DAY BAL | CHARGE SRC:DAY BAL | CHARGE SRC:DAY BAL
285	5.2	PROP SRC:DAY EQ WPMAX(100) | PROP SRC:DAY EQ WPMAX(100) | PROP SRC:DAY EQ WPMAX(100)
275	5.0	PROP SRC:B1 BAL WPMAX(100) | PROP SRC:B1 WAIT WPMAX(100) | PROP SRC:B1 WAIT WPMAX(100)
246	4.5	PROP SRC:B1 WAIT WPMAX(100) | PROP SRC:B1 BAL WPMAX(100) | PROP SRC:B1 BAL WPMAX(100)
242	4.4	PROP SRC:DAY EQ | PROP SRC:DAY EQ | PROP SRC:DAY EQ
236	4.3	CHARGE SRC:DAY EQ TOP CELL | FULL | FULL
161	3.0	PROP SRC:DAY BAL WPMAX(100) | PROP SRC:DAY WAIT WPMAX(100) | PROP SRC:DAY WAIT WPMAX(100)
133	2.4	PROP SRC:B1 EQ WPMAX(0) | PROP SRC:B1 EQ WPMAX(0) | PROP SRC:B1 EQ WPMAX(0)
126	2.3	CHARGE SRC:DAY EQ TOP | FULL | FULL
88	1.6	IDLE SRC:DAY EQ WPMAX(0) | IDLE SRC:DAY EQ WPMAX(0) | IDLE SRC:DAY EQ WPMAX(0)
78	1.4	IDLE SRC:DAY EQ | IDLE SRC:DAY EQ | IDLE SRC:DAY EQ
71	1.3	PROP SRC:STK WAIT | PROP SRC:STK BAL | PROP SRC:STK BAL
39	0.7	PROP SRC:DAY BAL | PROP SRC:DAY WAIT | PROP SRC:DAY WAIT
25	0.5	CHARGE SRC:DAY BAL TOP CELL | FULL | FULL
24	0.4	PROP SRC:DAY EQ WPMAX(0) | PROP SRC:DAY EQ WPMAX(0) | PROP SRC:DAY EQ WPMAX(0)
21	0.4	CHARGE SRC:DAY EQ CELL | FULL | FULL
19	0.3	PROP SRC:STK BAL | PROP SRC:STK WAIT | PROP SRC:STK WAIT
16	0.3	CHARGING | PROP SRC:DAY EQ WPMAX(100) | PROP SRC:DAY EQ WPMAX(100)
12	0.2	CHARGE SRC:DAY EQ | FULL | FULL
11	0.2	IDLE SRC:DAY EQ WPMAX(100) | IDLE SRC:DAY EQ WPMAX(100) | IDLE SRC:DAY EQ WPMAX(100)
10	0.2	CHARGE SRC:DAY EQ CCL | CHARGE SRC:DAY EQ | CHARGE SRC:DAY EQ
9	0.2	IDLE SRC:DAY WAIT WPMAX(100) | IDLE SRC:DAY BAL WPMAX(100) | IDLE SRC:DAY BAL WPMAX(100)
7	0.1	CHARGING | PROP SRC:B1 EQ WPMAX(100) | PROP SRC:B1 EQ WPMAX(100)
7	0.1	CHARGE SRC:DAY BAL CCL | FULL | FULL
5	0.1	PROP SRC:B1 EQ WPMAX(100) | CHARGING | CHARGING
5	0.1	IDLE SRC:DAY BAL WPMAX(100) | IDLE SRC:DAY WAIT WPMAX(100) | IDLE SRC:DAY WAIT WPMAX(100)
4	0.1	IDLE SRC:DAY BAL | CHARGING | CHARGING
4	0.1	CHARGING | IDLE SRC:DAY EQ | IDLE SRC:DAY EQ
4	0.1	PROP SRC:DAY WAIT WPMAX(100) | PROP SRC:DAY BAL WPMAX(100) | PROP SRC:DAY BAL WPMAX(100)
4	0.1	CHARGE SRC:DAY EQ CCL | CHARGE SRC:DAY EQ CCL | CHARGE SRC:DAY EQ CCL
3	0.1	PROP SRC:DAY BAL | CHARGING | CHARGING
3	0.1	CHARGE SRC:DAY BAL TOP CCL | FULL | FULL
3	0.1	CHARGE SRC:DAY EQ TOP CCL | FULL | FULL
3	0.1	CHARGE SRC:DAY EQ TOP CCL | CHARGE SRC:DAY EQ CCL | CHARGE SRC:DAY EQ CCL
2	0.0	IDLE SRC:STK EQ | IDLE SRC:STK EQ | IDLE SRC:STK EQ
2	0.0	CHARGING | PROP SRC:DAY EQ | PROP SRC:DAY EQ
2	0.0	CHARGE SRC:DAY BAL CCL | CHARGE SRC:DAY BAL | CHARGE SRC:DAY BAL
2	0.0	IDLE SRC:B1 EQ WPMAX(0) | IDLE SRC:B1 EQ WPMAX(0) | IDLE SRC:B1 EQ WPMAX(0)
2	0.0	IDLE SRC:STK WAIT | IDLE SRC:STK BAL | IDLE SRC:STK BAL
2	0.0	IDLE SRC:DAY BAL | IDLE SRC:DAY WAIT | IDLE SRC:DAY WAIT
2	0.0	PROP SRC:B1 WAIT WPMAX(0) | PROP SRC:B1 BAL WPMAX(0) | PROP SRC:B1 BAL WPMAX(0)
1	0.0	PROP SRC:DAY BAL WPMAX(100) | CHARGING | CHARGING
1	0.0	CHARGE SRC:STK EQ | CHARGE SRC:STK EQ | CHARGE SRC:STK EQ
1	0.0	IDLE SRC:DAY EQ WPMAX(0) | CHARGING | CHARGING
1	0.0	CHARGING | PROP SRC:DAY EQ WPMAX(0) | PROP SRC:DAY EQ WPMAX(0)
1	0.0	PROP SRC:DAY EQ WPMAX(100) | CHARGING | CHARGING
1	0.0	IDLE SRC:DAY WAIT WPMAX(0) | IDLE SRC:DAY BAL WPMAX(0) | IDLE SRC:DAY BAL WPMAX(0)
1	0.0	CHARGING | CHARGING | CHARGING
1	0.0	IDLE SRC:DAY EQ | CHARGING | CHARGING
1	0.0	PROP SRC:DAY EQ | CHARGING | CHARGING